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Hybrid Micro Grid Power System Technical and Economic Analysis Using Homer Legacy Software

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Research Article

Abstract

Inadequate capacity of Nigerian grid system force consumers of electricity to search for affordable alternative source of energy. This paper studies load pattern of Department of Electrical Engineering of Ahmadu Bello University, Zaria with an aim of finding out economic viability among different alternative sources available at the department. The energy sources were solar Photo Voltaic (PV), generator, battery bank and grid supply. Power and Harmonic Analyzer (PHA) was used to determine the load pattern of the department at diverse hours of days and months of a year. Subsequently, HOMER software was used to determine the best arrangement of the proposed system. Simulation results show that the solar/grid/diesel generator system with storage configuration is the most appropriate formation based on Net Present Cost (NPC) Approach which was found to be N38, 415,556:43 for projection of 25years operation period providing 43.6% and 44.66% savings as compared with hybrid PV/ diesel with battery backup and Hybrid PV/ diesel without battery storage respectively.

Keywords

Hybrid power system, NPC, HOMER, PHA

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1. Introduction

Nigeria is still battling with insufficient generation, transmission lines degradation and deficient distribution networks that will cater for the need of its populace (Oladipo *et al.*, 2018). So far, the national grid is not very reliable. Uninterruptable power supply is a basic requirement for the smooth running of any formal organization. Different energy resources are required to ensure qualitative and constant power supply at affordable cost. In hybridizing different resources, the best technical configuration and economic viability need to be investigated (Shaahid & Elhadidy, 2007). Solar energy is considered as clean and cheaper source that can be easily harness to improve reliability of supply, this is possible because it is free from carbon emission and absent of sounds associated with diesel generator. However, it has a shortcoming: the sun can only be available during the day time (Das *et al.*, 2016; Pradhan *et al.*, 2017). Generally, Renewable Energy Resources (RER) are characterized with high installation cost and a very low running cost, whereas reverse is the case for conventional energy resources. HOMER legacy software is an optimizer that requires some basic information about a system. It runs, computes and provides results of different possible configurations and economic implications using Total Net Present Cost (TNPC) analysis aims at choosing the best, technically, and economical viable one (Usman *et al.*, 2018). HOMER was brought into being by National Renewable Energy Laboratory (NREL) of United State of America as general purpose software for grid and standalone power system generation (Das *et al.*, 2016; Kishore & Kumar, 2019; Papaioannou *et al.*, 2014; Shahzad *et al.*, 2017) The software needs a load profile of a propose system, cost of different materials and weather data such as solar irradiance of an area of consideration, temperature, wind speed etc. as the case may be.

A microgrid is a restricted cluster of energy sources and electric loads that can work either connected to the wide area interconnected power grid (macro grid) or as a standalone (Papaioannou *et al.*, 2014). The idea of microgrid is to facilitate easy control, accessibility, and improve system reliability. One of the major questions in formation of microgrid system is how cheap the system is, in this paper, system configurations are setup and the cheapest are computed and selected. The paper is organized in four sections; the introduction is the first section, followed by methodology in the second section. Results with their respective discussions are contained in section three. Finally, conclusion is set in section number four.

2. Methodology

Sizing of a microgrid system in this approach require two important stages for efficient analysis of energy resources. The stages include: determination of load profile (i.e. the consumption) and estimation of materials, production, and cost stages.

2.1 Load profiling

The power and Harmonic Analyzer (PHA) is an instrument used to measure a real time power system parameters (line and Phase Voltages, currents, Real, Apparent and Reactive power, power factor, and so on). PHA was used for load profiling: Measurement was taken using the device at an interval of 10min for 30 days which enable the choice of the peak load demand within the test period. Figure 1 shows set up of PHA connected to the mains to enable measurements of the total power consumed at the Department of Electrical Engineering.

Figure 1: Measurement setup

2.2 Estimation of Materials, Production and Cost of *HOMER software*

In order to investigate a suitable configuration, system optimization and analysis was carried out using widely used software known as HOMER Pro Edition v64.3.11.2.The HOMER is a renewable energy optimization software device which is use to evaluate both grid-connected and standalone power systems for a variety range of applications. It operates on hourly data for a year and computes energy flow with cost implications. A simulation of three most likely system provisions with various system architecture options based on NPC was done for evaluation. Among the important features of HOMER is sensitivity analysis; this iterations, and the optimization procedure for every sensitivity variable indicated by the user. The analysis shows how the outputs response according to inputs variation.

Choices of variables used in this work are: Size of PV array, size of Generator, number and size of batteries, Converter size and grid power price.

PV array

Power supplied by PV was obtained from Equation (1).

$$P_{PV} = Y_{PV}\mathcal{F}_{PV}\left(\frac{G_T}{G_T,STC}\right)\left[1+\alpha_p(T_c - T_c,STC)\right] \quad (1)$$

Where:

Y_{PV} = Photovoltaic array rated capacity, or PV output power for standard test situations [kW];

$\mathcal{F}_{PV}$ = PV derating factor [%];

G_T = solar radiation incident on the PV array [kW/m^2];

G_T,STC = incident radiation at standard test settings,1 kW/m^2];

α_p = temperature coefficient of power [%$^\circ$C];

T_c = PV cell temperature in the current time step [$^\circ$C];

T_c,STC = PV cell temperature under normal test environment [25°C].

Diesel generator

Due to their uncertainty and intermittency of renewable energy sources, diesel generator functions as standby power source to deliver continuous power supply and also charge back-up battery. Essentially, energy balance is requiring for optimal system operation as consumption of fuel is a function of power being supplied by the Diesel generator. Fuel consumed by diesel generator (F_G in litre/h), is modeled as dependent on the Diesel generator power provided.

$$F_G = B_G \times P_{G-rated} + A_G \times P_{G-out} \quad (2)$$

Where:

$P_{G-rated}$ *is the nominal power of the diesel generator*

P_{G-out} is the output power

A_G *and* B_G are the coefficients of fuel consumption curve as define by the user /KWh

Storage

Battery was used as storage device which serves as a backup when there is shortage in supply from energy sources. In this work lead-acid battery was chosen for the design. Battery being a non-linear device when modeling it in real time analysis some key parameters need to be considered.

1. Battery state of Charge
2. Rate of charge/discharge
3. Ambient temperature
4. Life time and other phenomenon, such as gassing, self-discharge and heating loss.

Storage capacity of the battery was calculated using the following expression.

$$C_{Wh} = {(E_L \times AD)}\Big/{(\eta_{inv} \times \eta_{Batt} \times DOD)} \quad (3)$$

Where:

E_L is the average daily load energy (kWh/day)

AD is the days of Autonomy of the system

DOD is the battery Depth of Discharge

η_{inv} *and* η_{Batt} are the inverter and Battery efficiencies

Evaluation of system cost

NPC is an existing worth of all the system charges acquired over its entire operation time; subtract the present value of all the income brought over its lifetime. The Costs include setting cost, Operation and Maintenance costs otherwise call running coast, replacement costs, fuel expenses, discharges consequences, and costs of grid power. Revenues include: salvage value and grid transactions. System NPC is calculated using the Equation (4).

$$C_{NPC} = \frac{TAC}{CRF(i,N)} \quad (4)$$

Where:

$$CRF(i,N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (5)$$

HOMER calculates a levelized Cost of Energy (COE) as the average cost per kWh of useful electrical energy produced by the system. Equation 6 was used to calculate COE of the system.

$$COE = \frac{C_{ann,tot}}{E_{prim,AC}} \quad (6)$$

Where:

$C_{ann,tot}$ = total annualized cost of the system [#/yr.];

$E_{prim,AC}$ = AC primary load served [kWh/yr].

The excess electricity fraction is a ratio of total excess electricity to total electricity produce. HOMER utilizes equation 7 to determine the value of excess electricity at the end of each simulation.

$$f_{excess} = \frac{E_{excess}}{E_{prod}} \tag{7}$$

Where:
E_{excess} is the total excess electricity [$KWhr/yr$]
E_{prod} is the total electricity produce [$KWhr$/yr]

3. Results and Discussion

3.1 Load Profile

After a careful study of load profile of the department carried out with PHA, a load pattern of Figure 2 was obtained as the maximum demand recorded over the one-month measurements. The PHA data is assumed to be a peak demand because it was captured during the full study season for both undergraduate and postgraduate students. The data capturing therefore, was spread over the year.

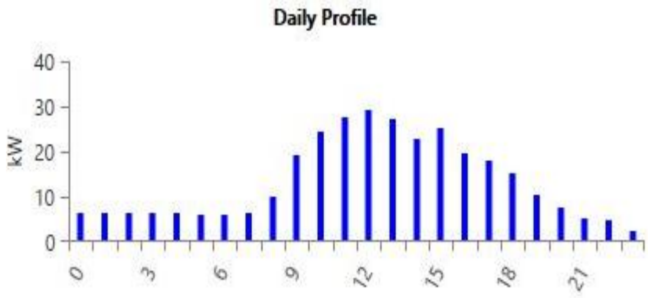


Figure 2: Maximum daily load profile

The load profile indicates a maximum close to 30kW at mid of a day and a minimum of around 2.5 kW at the night hours as observed in Figure 2. This indicates that the 5kW PV supply cannot cover the peak hour as it represents only 16.67% of the demand. Although, it can cover the off-peak hours with even excess energy that can be sell to grid.

3.2 Economic Aspect

Data of solar irradiance, cost of grid energy, cost of diesel, cost of generators, cost of battery, charge controller as well as cost PV modules were all obtained and successfully implemented to the software, simulations was carried out and the following results were obtained.

Scenario I: *Hybrid Photovoltaic (PV)/Diesel System without storage.*

Hybrid PV/Diesel system without battery combination was simulated and the result indicates a total NPC of N 88,107,870.00. The power generated by PV is being completely consumed during the day when the demand is highest. However, the surplus solar power, which could have been used to charge the battery (which was not available in this case), was considered as loss. As a result, the PV cannot meet the demand when the sun sets, since there were no storage elements that exist to cover the demand in the evening and the early hours of morning figure 3 shows the systematic diagram of the system. Generator will come ON in order to reach with the demand. Figure 4 shows details of the analysis.

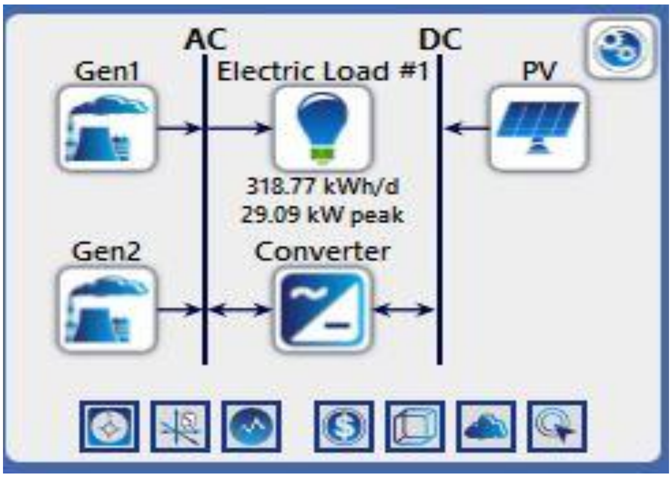


Figure 3: Systematic diagram of scenario I

Component	Capital (₦)	Replacement (₦)	O&M (₦)	Fuel (₦)	Salvage (₦)	Total (₦)
GEN 1	₦1,800,000.00	₦8,175,195.54	₦6,401,706.19	₦33,755,718.86	(₦265,190.54)	₦49,867,430.06
GEN 2	₦2,700,000.00	₦2,328,485.32	₦2,596,879.52	₦16,282,001.05	(₦122,246.37)	₦23,785,119.52
Generic flat plate PV	₦7,753,840.00	₦0.00	₦1,861,562.38	₦0.00	₦0.00	₦9,615,402.38
System Converter	₦3,600,000.00	₦1,527,385.79	₦0.00	₦0.00	(₦287,469.42)	₦4,839,916.37
System	₦15,853,840.00	₦12,031,066.65	₦10,860,148.10	₦50,037,719.91	(₦674,906.33)	₦88,107,868.33

Figure 4: Total NPC for Hybrid PV/Diesel System without battery simulation outcome

Scenario II: *Hybrid PV/Diesel with batteries.*

The system configuration can be seen in figure 5, in the simulation outcome, hybrid PV, generator and battery storage has a total NPC of N86, 023,837.72 Figure 6 indicates the scenario's simulation result. This was found to be cheaper than hybrid PV/diesel configuration without battery projected in the preceding scenario because, excess solar power was stored for utilization during the night hours.

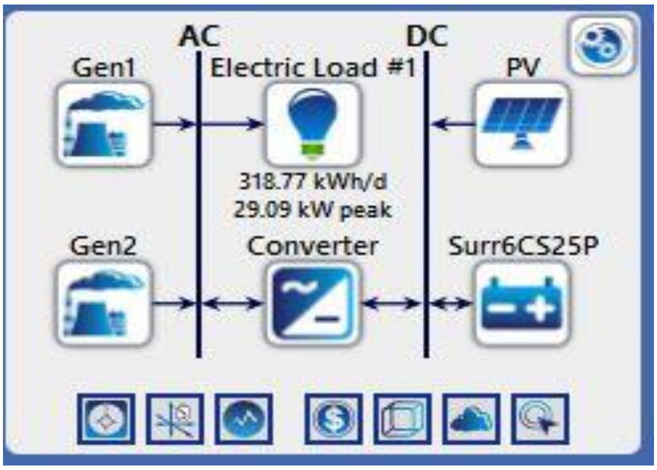
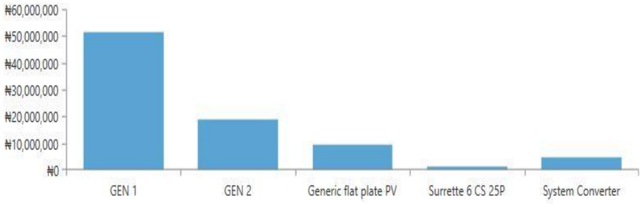


Figure 5: Systematic Diagram of scenario II



Component	Capital (₦)	Replacement (₦)	O&M (₦)	Fuel (₦)	Salvage (₦)	Total (₦)
GEN 1	₦1,800,000.00	₦8,161,513.63	₦6,384,124.77	₦35,290,675.94	(₦276,186.25)	₦51,360,128.10
GEN 2	₦2,700,000.00	₦1,182,950.79	₦1,847,600.67	₦13,302,411.24	(₦8,731.88)	₦19,024,230.81
Generic flat plate PV	₦7,753,840.00	₦0.00	₦1,861,562.38	₦0.00	₦0.00	₦9,615,402.38
Surrette 6 CS 25P	₦830,000.00	₦383,647.20	₦0.00	₦0.00	(₦29,487.15)	₦1,184,160.05
System Converter	₦3,600,000.00	₦1,527,385.79	₦0.00	₦0.00	(₦287,469.42)	₦4,839,916.37
System	₦16,683,840.00	₦11,255,497.41	₦10,093,287.82	₦48,593,087.18	(₦601,874.70)	₦86,023,837.72

Figure 6: Hybrid PV/diesel with Storage simulation result

Scenario III: *Grid connected Hybrid PV/Diesel with storage.*

Grid connected Hybrid PV/Diesel with storage are configured as seen in figure7. In this occasion, the total NPC for the system was N38, 415,556:43. The NPC here was very much lower than the two scenarios considered earlier. Figure 8 presented the simulation result of the scenario. The result generated a lower NPC because of the present of grid supply which was very much cheaper than all other sources, in addition storage was considered in this case hence excess energy during the day time was not wasted.

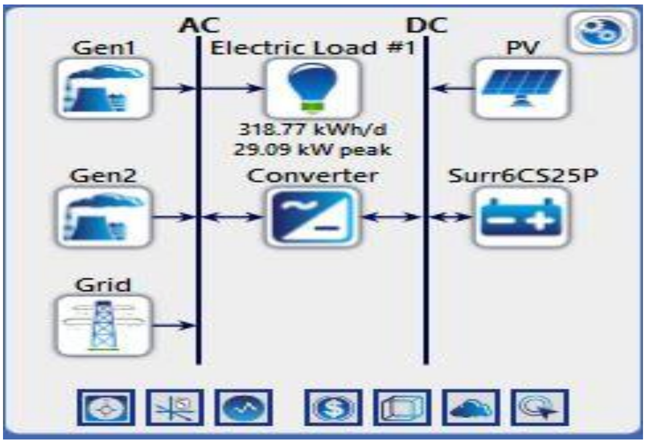
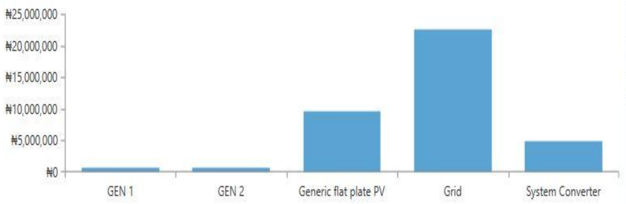


Figure 7: Systematic diagram of scenario III



Component	Capital (₦)	Replacement (₦)	O&M (₦)	Fuel (₦)	Salvage (₦)	Total (₦)
GEN 1	₦900,000.00	₦0.00	₦0.00	₦0.00	(₦189,190.81)	₦710,809.19
GEN 2	₦900,000.00	₦0.00	₦0.00	₦0.00	(₦189,190.81)	₦710,809.19
Generic flat plate PV	₦7,753,840.00	₦0.00	₦1,861,562.38	₦0.00	₦0.00	₦9,615,402.38
Grid	₦0.00	₦0.00	₦22,538,619.31	₦0.00	₦0.00	₦22,538,619.31
System Converter	₦3,600,000.00	₦1,527,385.79	₦0.00	₦0.00	(₦287,469.42)	₦4,839,916.37
System	₦13,153,840.00	₦1,527,385.79	₦24,400,181.69	₦0.00	(₦665,851.04)	₦38,415,556.43

Figure 8: Grid-connected Hybrid PV/Diesel with storage result

4. Conclusion

Grid power supply happens to be cheaper than PV and diesel generator with or without storage. However, its availability is questionable in Nigeria of today. Investigations indicate that PV/grid/diesel generator system with storage configuration is the most suitable configuration based NPC Approach. N38, 415,556:43 was obtained as NPC for 25years operation period providing 43.6% and 44.66 % saving for the other scenarios in which grid supply was not considered, leading to a total cost reduction of #49,692,311.90 and #47,608,281.29 respectively. Hence despite the problems of grid supply, PV System and diesel Generator cannot replace grid, can only serves as backups.

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